

CHAPTER 3: **PROFESSIONAL ETHICS**

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OBJECTIVES

- Describe the relationship to be managed by IT professional
- State the professional code of ethics for software engineers
- Know the related law for IT
- Making an ethical decision

CONTENTS

- IT professional
- IT professional relationship to be managed
- Professional Code of Ethics
- Related laws in regards of IT/ICT
- Making ethical decision

IT PROFESSION

- Profession (Webster) : A calling requiring specialized knowledge and often long and intensive academic preparation
- Profession (informal)
 - High level of education
 - Practical experience
 - Decisions have impact

IT PROFESSION

- **Professional:** someone who requires advanced training and experience, must exercise discretion and judgment in the course of his or her work
- **Who recognizes his or her obligation to society by living up to established and accepted codes of conduct**



IT PROFESSIONAL



- Many workers in IT industry are considered to be professional:
- programmers/analysts
- Software engineers
- Database administrators
- LAN administrators
- CIOs
- etc



PROFESSIONAL RELATIONSHIPS TO BE MANAGED

- An ethical IT professional acts honestly and appropriately, and
- at all time steer away from any unethical action
- Relationships
 - IT professional-employer
 - IT professional-client
 - IT professional-supplier
 - IT professional-professional
 - IT professional-IT user
 - IT professional-society



IT PROFESSIONAL-EMPLOYER

- **Term of employment**
 - job title, performance expectation, work responsibility, drug testing, location of employment, salary, dress code, etc.
- **Policies, procedures manual, or code of conduct**
 - protection of secrets, vacations, use of company resources, etc
- **Some specific aspect – programming language to be used, etc**

IT PROFESSIONAL-EMPLOYER

- Must set example and enforce the policies in regards to ethical use of IT
- Have knowledge and skill to abuse systems and data or to allow others to do so
- Some ethical issues
 - software piracy
 - trade secret
 - whistle-blowing

IT PROFESSIONAL-CLIENT

- Whom professional provides services
- Can be from outside or internal
- Some ethical issues
 - actual status of projects
 - trusted recommendation

IT PROFESSIONAL-SUPPLIER

- Dealing with hardware, software, and service providers
- Some ethical issues
 - fair in dealing with suppliers and do not make unreasonable demands
 - accept bribe

IT PROFESSIONAL-PROFESSIONAL

- Loyalty to other members of profession
- Some ethical issues
 - resume inflation
 - inappropriate sharing of corporate information

IT PROFESSIONAL-IT USER

- Employs the product to deliver organization benefits or to increase productivity
- A duty to understand to needs and capabilities of users
- To deliver product and services that best meet these needs within budget and time constraint
- Discourage piracy, minimize the appropriate use of computing resources, avoid the inappropriate sharing of information
- Establishment of IT usage policy

IT PROFESSIONAL-SOCIETY

- Practice in ways that not only causes no harm but also provide significant benefits



- To establish and maintain professional standards that protect public

CLASS DEBATE

How do you distinguish between **misrepresentation** and **embellishment** of one's professional accomplishments in order to win a contract to complete a major project? Provide an example of an embellishment that would not be considered misrepresentation.

PROFESSIONAL CODES OF ETHICS

- States the principles and core values essential to the work of a particular occupational group.
- Laws do not provide a complete guide to ethical behaviour – **illegality does not mean ethical**
- Help to ensure:
 - abiding the laws
 - following necessary regulations
 - behaving in an ethical manner

PROFESSIONAL CODES OF ETHICS - BENEFITS

- Improves ethical decision making
 - use common set of core values and beliefs to serve as a guideline
- Promote high standard of practice and ethical behavior
 - reminder of responsibilities and duties
 - defines acceptable and unacceptable behavior
- Enhances trust and respect from the public
 - depends on the integrity and good judgment to tell the truth, abstain from giving self-serving advice, and offer warning of potential negative side effects.
- Provides an evaluation benchmark
 - self assessment and peer evaluation

TYPES OF CODE OF ETHICS

- Association for Computing Machinery (ACM) Code of Ethics and Professional Conduct
- Association of Information Technology Professional (AITP) Code of Ethics
- Computer Society of the Institute of Electrical and Electronics Engineers (IEEE-CS) – Software Engineering Code of Ethics and Professional Practices (SECEPP)

SOFTWARE ENGINEERING CODES OF ETHICS AND PROFESSIONAL PRACTICE (SECEPP)

- Preamble

- Principles

- download at:

- <http://www.ics.uci.edu/~redmiles/ics131-FQ03/week08ethics/IEEE-ACM-Ethics.PDF>

PREAMBLE

- Software engineers have opportunities to do good or cause harm, or to influence others to do good or cause harm
- Software engineers ought to be committed to doing good
- Eight principles identify key ethical relationships and obligations within these relationship
- Code should be seen as a whole, not a collection of parts
- Concern for the public interest is paramount

PRINCIPLES

■ Principle 1 **Public**

- software engineers shall act consistently with the public interest

■ Principle 2 **Client and Employer**

- software engineers shall act in a manner that is in the best interests of their client and employer and that is consistent with the public interest

PRINCIPLES

■ Principle 3 **Product**

- software engineers shall ensure that their products and related modifications meet the highest professional standards possible

■ Principle 4 **Judgment**

- software engineers shall maintain integrity and independence in their professional judgment

PRINCIPLES

■ Principle 5 **Management**

- software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance

■ Principle 6 **Profession**

- software engineers shall advance the integrity and reputation of the profession consistent with the public interest

PRINCIPLES

■ Principle 7 **Colleagues**

- software engineers shall be fair to and supportive of their colleagues

■ Principle 8 **Self**

- software engineers shall participate in lifelong learning regarding the practice of their profession and promote an ethical approach to the practice of the profession

ALTERNATIVE LIST (VIRTUES DISCIPLINE-INDEPENDENT)

- Be impartial : 1.02, 1.03, 1.05, 1.07, 3.03, 3.12, 4.01, 6.05
- Disclose information that others ought to know : 1.04, 1.06, 2.06, 2.07, 3.01, 4.05, 4.06, 5.05, 5.06, 6.07, 6.08, 6.09, 6.12, 6.13
- Respect the rights of others : 2.02, 2.03, 2.05, 3.13
- Treat others justly : 5.06, 5.07, 5.08, 5.09, 5.10, 5.11, 5.12, 7.03, 7.04, 7.05, 7.07, 8.07

ALTERNATIVE LIST (VIRTUES DISCIPLINE-INDEPENDENT)

- Take responsibility for your actions and inaction :
1.01, 3.04, 3.05, 3.06, 3.07, 3.08, 3.10, 3.11,
3.14, 3.15, 4.02, 7.08
- Take responsibility for the actions of those you
supervise : 5.01, 5.02, 5.03, 5.04
- Maintain your integrity : 2.01, 2.04, 2.08, 2.09,
3.01, 3.02, 3.09, 4.03, 4.04, 6.06, 6.10, 6.11,
8.08, 8.09
- Continually improve your ability : 8.01-8.06
- Share your knowledge, expertise, and values:
1.08, 6.01-6.04, 7.01, 7.02, 7.06

CASE STUDY – EVALUATE MORAL PROBLEM USING SECEPP

■ A three step process

- 1) consult and identify relevant fundamental **principle** (alternative list)
- 2) search the list of **clauses** to see the most directly speak for the relevant principles
- 3) determine whether the **contemplated action aligns with or contradicts the statement** in the clauses. If the action agrees with all of the clauses --> strong evidence the action is moral. If is in disagreement with all the clauses --> safe to say the action is immoral

CASE 1– SOFTWARE RECOMMENDATION

SCENARIO

Sam Shaw calls the Department of Computer Science at East Dakota State University seeking advice on how to improve the security of his business's local area network. A secretary in the department routes Mr. Shaw's call to Professor Jane Smith, an internationally recognized expert in the field. Professor Smith answers several questions posed by Mr. Shaw regarding network security. When Mr. Shaw asks Professor Smith to recommend a software package to identify security problems, Professor Smith tells him that NetCheks got the personal computer magazine's top rating. She does not mention that the same magazine gave a "best buy" rating to another product with fewer features but a much lower price. She also fails to mention that NetCheks is a product of a spin-off company started by one of her former students and that she owns 10 percent of the company.

CASE 1– SOFTWARE RECOMMENDATION

- Sam Shaw asks for free advise on LAN security
- Prof. Smith answers questions and recommends top-ranked package
- Prof Smith does not disclose
 - she has financial interest in company producing top-ranked package
 - another package was given a “best-buy” rating
- Did Prof. Smith do anything wrong?

ANALYSIS – SOFTWARE RECOMMENDATION

- Most relevant **principles/disciplines**
 - a) be impartial
 - b) disclose information others ought to know
 - c) share your knowledge, expertise, and values
- **Clause** 1.06: Prof Smith was deceptive
- **Clauses** 1.08, 6.02: prof smith freely gave valuable information
- **Clauses** 4.05,6.05 : prof smith did not reveal conflict of interest
- **Conclusion:** Professor Smith should have revealed her conflict of interest to Mr. Shaw.

CASE 2: CHILD PORNOGRAPHY

Joe Green, a system administrator for a large corporation, is installing a new software package on the PC used by employee Chuck Dennis. The company has not authorized Joe to read other people's emails, Web logs, or personal files. However, in the course of installing the software, he accidentally comes across directories containing files with suspicious-looking names. He opens a few of

files and discovers they contain child pornography. Joe believes possessing the images is against federal law. What should he do?

CASE 2: CHILD PORNOGRAPHY

- Joe Green a system administrator
- Asked to install new software package on Chuck Dennis's computer
- Green not authorized to read other people's emails or personal files
- Green sees suspicious-looking file names
- He opens some of Dennis's files and discovers child pornography
- What should he do?

ANALYSIS (1/2)

- Most relevant **principles/disciplines**
 - Be impartial
 - Respect the rights of others
 - Treat others justly
 - Maintain your integrity

ANALYSIS (2/2)

- Most relevant **clauses**
 - 2.03: Somebody has misused the company PC
 - 2.09: Someone is using the PC for a purpose not in the employer's interest
 - 3.13: Joe violated the policy against opening files
 - 5.10: Someone else may have planted the files on Chuck's computer

CONCLUSIONS

Conclusions:

- Joe was wrong to violate company policy to uncover child pornography
- Once he has this knowledge, however, he is obliged to share it with company authorities
- Joe should be discreet

CASE 3: ANTI-WORM

- Internet plagued by new worm that exploits hole in popular operating system
- Tim Smart creates anti-worm that exploits same security hole
- Tim's anti-worm fixes PCs it infects. It also uses these PCs as launch pad to reach new PCs.
- Tim launches anti-worm, taking pains to keep it from being traced back to him.
- The anti-worm quickly spreads through Internet, infecting millions of computers
- System administrators around the world combat the anti-worm

ANALYSIS (1/2)

- Most relevant **principles**
 - Continually improve your abilities.
 - Share your knowledge, expertise, and values.
 - Respect the rights of others.
 - Take responsibility for your actions and inactions.

ANALYSIS (2/2)

■ Most relevant **clauses**:

- 1.01: Tim did not accept responsibility for his action.
- 1.08: The worm was free, but cost system administrators a lot of time.
- 2.03: The anti-worm entered computers without permission of their owners.
- 8.01, 8.02, 8.06: Tim improved his knowledge and skills by creating the anti-worm.

CONCLUSIONS

Conclusions:

- Tim's welfare is less important than the public good
- By attempting to hide his identity, Tim refused to accept responsibility for his actions
- Tim violated the property rights of the PC owners whose systems were infected by his anti-worm
- Tim violated the Code

CASE 4: CONSULTING OPPORTUNITY

- Jean works in support organization for Acme Corporation
- Many Acme customers downgrading their level of support
- East Dakota gives Jean opportunity to run a training class similar to that provided by Acme
- Jean tells no one at Acme
- Jean develops materials at home on own time
- Jean takes paid vacation to teach class

ANALYSIS (1/2)

- Most relevant **principles**
 - Be impartial.
 - Take responsibility for your actions and inactions.
 - Disclose information that others ought to know.
 - Maintain your integrity.
 - Continually improve your abilities.

ANALYSIS (2/2)

- Most relevant **clauses:**

- 3.04: Jean was well qualified to develop materials and teach class
- 8.04: By creating materials, Jean became even more familiar with Acme's package and its capabilities
- 4.05: Jean didn't disclose his conflict of interest with his employer
- 2.08: Jean deprived himself of "time off" needed to do his best work at Acme
- 6.05: Jean put his own interest above that of his employer

CONCLUSIONS

Conclusions:

- Jean did not disclose East Dakota's offer or his decision to Acme's management
- Acme's management is likely to question Jean's loyalty to the company
- Jean's actions were wrong and unwise

ACM CODE OF ETHICS

- Consisting of 24 imperatives formulated as statements of personal responsibility, identifies the elements of such a commitment
- General Moral Imperatives
- Specific Professional Responsibility
- Organizational Leadership Imperatives
- Compliance with the Code

GENERAL MORAL IMPERATIVES

As an ACM member I will:

- 1.1 Contribute to society and human well-being
- 1.2 Avoid harm to others
- 1.3 Be honest and trustworthy
- 1.4 Be fair, and take action not to discriminate
- 1.5 Honor property rights including copyrights and patents
- 1.6 Give proper credit for intellectual property
- 1.7 Respect privacy of others
- 1.8 Honor confidentiality

SPECIFIC PROFESSIONAL RESPONSIBILITIES

As an ACM computing professional I will:

- 2.1 Strive to achieve the highest quality in both the process and products of professional work
- 2.2 Acquire and maintain professional competence
- 2.3 Know and respect existing laws pertaining to professional work
- 2.4 Accept and provide appropriate professional review
- 2.5 Give comprehensive and thorough evaluation of computer system and their impacts, including analysis of possible risk

SPECIFIC PROFESSIONAL RESPONSIBILITIES

As an ACM computing professional I will:

2.6 Honor contracts, agreements, and assigned responsibilities

2.7 Improve public understanding of computing and its consequences

2.8 Access computing and communication resources only when authorized to do so

ORGANIZATIONAL LEADERSHIP IMPERATIVES

As an ACM member and an organizational leader, I will:

- 3.1 Articulate social responsibilities of members of an organizational unit and encourage full acceptance of those responsibilities
- 3.2 Manage personnel and resources to design and build information systems that enhance the quality, effectiveness and dignity of working life
- 3.3 Acknowledge and support proper and authorized uses of an organization's computing and communications resources

ORGANIZATIONAL LEADERSHIP IMPERATIVES

As an ACM member and an organizational leader, I will:

- 3.4 Ensure that users and those who will be affected by a computing system have their needs clearly articulated during the assessment and design of requirement. Later the system must be validated to meet requirements
- 3.5 Articulate and support policies that protect the dignity of users and others affected by a computing system
- 3.6 Create opportunities for members of the organization to learn the principles and limitations of computer systems

COMPLIANCE WITH CODE

- **4.1 Uphold and promote the principles of this code**
- **4.2 Treat violations of this code as inconsistent with membership in the ACM**

SOME EMPHASIZES OF THE CODES

- Honesty and fairness

SE 1.6, 2.1, 6.7, Professional behaviour, 7.4, 7.5

ACM 1.3, 1.4

responsibilities to respect confidentiality

SE 2.5, ACM 1.8

- Maintain professional competence

SE 8.1 – 8.5, AM 2.2

- Aware of relevant laws

SE 8.5, ACM 2.3

- Honor contracts and agreement

ACM 2.6

SOME EMPHASIZES OF THE CODES

- **Respect and protect privacy**
SE 1.3, 3.12, ACM 1.7
- **Avoid harm to others**
SE 1.3, ACM 1.2
- **Respect property right**
SE 2.2, 2.3, ACM 1.5, 1.6, 2.8

IT USERS

- Software piracy.
- Inappropriate use of computing resources.
- Inappropriate sharing of information.

SUPPORTING ETHICAL PRACTICES OF IT USERS

- **Define and limit the appropriate use of IT resources.**
- **Establish guidelines for the use of company software.**
- **Structure information systems to protect data and information.**
- **Install and maintain a corporate firewall**

SUMMARY

- A professional is someone:
 - who requires advanced training and experience.
 - who exercises discretion and judgment during work.
 - whose work cannot be standardized.
- IT professionals have many different relationships that have different ethical issues.

SUMMARY

- A professional code of ethics states the principles and core values essential to the work of a particular occupational group.
- Licensing and certification of IT professionals would increase the reliability and effectiveness of information systems.

SUMMARY

- IT users encounter many ethical issues, including:
 - Software piracy.
 - Inappropriate use of IT resources.
 - Inappropriate sharing of private and secret data.
- An IT usage policy helps users understand how to appropriately use IT resources.